

COMP1549 – Advanced Programming

Lab 1: Programming in Java

Thursday, January 15, 2026

In Eclipse, create a Java project, and then create a package with the name **uk.ac.gre.comp1549.examples**. Later copy the files from **Demo.zip** into the created package.

Task 1

- Change the main method of Gravity class so that it computes the position of an object after falling for 10 seconds, outputting the position in meters. The formula in Math notation is:

$$x(t) = 0.5 \times at^2 + v_i t + x_i$$

Task 2

1. Write a method in Marathon class that takes as input an array of integers and returns the index corresponding to the person with the lowest time. Run this method on the array of times. Print out the name and time corresponding to the returned index.
2. Write a second method in Marathon class to find the second-best runner. The second method should use the first method to determine the best runner and then loop through all values to find the second-best (second lowest) time.

Task 3

1. Implement the methods in Book class and run the main method to see, if it works as expected
2. Add implementation in Library class that will represent each library and manage a collection of books. All libraries have the same hours: 9 AM to 5 PM daily. However, they have different addresses and book collections (i.e., arrays of Book objects). You must not change the main method.

The current class provides a main method that creates two libraries, then performs some operations on the books. However, all the methods and member variables are missing. You will need to define and implement the missing methods. Read the main method and look at the compile errors to figure out what methods are missing.